A short combinatorial proof of Seymour's second-neighbourhood conjecture

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Abstract

We give a short, self-contained and purely combinatorial proof of Seymour's conjecture on second neighbourhoods. Every nonempty finite oriented graph has a vertex with at least as many vertices at directed distance exactly two as at directed distance one. The main ingredient is a new deletion lemma for two families of ordered pairs. We prove it by induction, using Hall's defect theorem to reduce the instance and, when both Hall matchings exist, a combination of greedy matchings and double counting. The lemma applies to arbitrary instances on four finite sets and bounds the deletion cost by the smaller of two matching numbers, plus the number of points left uncovered. We also derive a weighted form of the conjecture and a consequence for directed triangles.

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1 Statement and plan

An *oriented graph* is a finite directed graph without loops and without a pair of opposite arcs. For a vertex v let $N_1^+(v) = \{u : v \to u\}$ and let $N_2^+(v)$ be the set of vertices at directed distance exactly two from v. For a set S of vertices write

$$F(S) = \{y : \ x \to y \text{ for some } x \in S\}, \qquad F^2 = F \circ F.$$

We allow $F(S)$ to meet S. Since an oriented graph has no directed closed walk of length two, $N_2^+(v) = F^2(\{v\}) \setminus F(\{v\})$.

Theorem 1.1 (Seymour's second-neighbourhood conjecture)**.** *Every nonempty oriented graph has a vertex v with $|N_1^+(v)| \le |N_2^+(v)|$.*

The conjecture was recorded by Dean and Latka [3], together with its special case for tournaments; that case was proved by Fisher [7] and, by a different method, by Havet and Thomassé [10]. Fidler and Yuster [6] proved the conjecture for several further classes of dense orientations, among them tournaments missing a star and tournaments missing a subtournament, and Ghazal [8] proved a weighted version for tournaments missing a generalized star. Kaneko and Locke [12] settled oriented graphs of minimum outdegree at most six. For arbitrary oriented graphs, Chen, Shen and Yuster [2] found a vertex v with $|N_2^+(v)| \ge \gamma |N_1^+(v)|$ for a constant $\gamma \approx 0.657$, and Huang and Peng [11] raised the constant to about 0.7155. Espuny Díaz, Girão, Granet and Kronenberg [5] showed that for every fixed $p < 1/2$, asymptotically almost surely every orientation of the random

graph $G(n,p)$ satisfies the conjecture. A proof of the full conjecture was released by OpenAI in 2026 [13]. Call a vertex *good* if it satisfies the inequality of Theorem 1.1 and *bad* otherwise. The proof has three steps.

(1) A minimal counterexample D satisfies a strict subset deficit (Lemma 2.1).

(2) A lexicographic blow-up of D is an oriented graph G with positive indegrees in which $|U| + |F^2U| < 2|FU|$ for every nonempty proper vertex set U (Lemma 2.2).

(3) No such G exists. This is the following set version of the conjecture, which holds for every oriented graph with positive indegrees, not only for blow-ups.

Theorem 1.2 (set version)**.** *Every nonempty oriented graph in which every vertex has an in-neighbour has a nonempty proper vertex set U with $|U| + |F^2(U)| \geq 2|F(U)|$.*

We prove Theorem 1.2 in Section 3 by an extremal argument based on a deletion lemma (Theorem 4.12). The reduction uses the minimal-counterexample and graph-product methods of Brantner, Brockman, Kay and Snively [1] and Seacrest's subset perspective [15]; we follow the presentation of these ideas in [13].

Our main new contribution is the purely combinatorial proof of the deletion lemma in Section 4. The first proof obtains the deletion bound through an auxiliary bipartite graph on copies of the Z-points, using Kőnig's theorem, an augmenting-path exchange, Edmonds' generic-rank correspondence and a kernel lemma. Here we prove the lemma directly: Hall's defect theorem handles the inductive cases, while greedy matchings and a double-counting argument handle the saturated case. The theorem applies to arbitrary instances on four finite sets and gives a sharper bound in terms of the smaller of two matching numbers; the weaker bound $|H| + q$ suffices for Theorem 1.2. Section 5 records two consequences, and Section 6 describes the formal verification in Lean.

Outline

In a counterexample with as few vertices and then as few arcs as possible, every nonempty proper set S satisfies

$$|F^2S \setminus FS| < |FS \setminus S|$$

(Lemma 2.1): deleting carefully chosen arcs produces a good vertex, and the neighbourhoods it loses force this deficit. Replacing each vertex by a transitive tournament on $m > n$ vertices magnifies the deficit by m, while each fibre contributes an error of at most one; this yields the strict concavity $|U| + |F^2U| < 2|FU|$ (Lemma 2.2).

To rule out concavity (Section 3), we consider pairs (P, Q) of sets of ordered pairs of vertices that are *compatible*: the images under F of the columns of P and of the rows of Q are disjoint. Among them we maximize $|P| + |Q| + d$, where d counts the points outside both images. Summing concavity over the columns of P and the rows of Q produces new families R, C whose combined size exceeds $|P| + |Q|$ by more than $2d$. These families need not be compatible, but the deletion lemma (Theorem 4.12) restores compatibility at a cost of at most $d + q$, where q counts the points that the repaired families leave uncovered. These q points are credited back to the objective, which therefore increases.

The deletion lemma is proved by induction (Section 4). Two conflicting members determine an H-point, and the conflicts with a fixed H-point form a complete bipartite block, so we pay one unit per H-point. If the H-points cannot be matched into one of the families, Hall's theorem gives a deficient set of H-points with few neighbours there; deleting these few members removes

all those H-points, and induction applies. If both matchings exist, we delete everything. The counting lemma (Lemma 4.11) bounds the cost: each family can be matched into the Z-points it covers (Lemma 4.10), and the two matchings have total size at most $|Z| + |H|$ (Lemma 4.9). This matching inequality is the combinatorial shadow of a factorization through H (Remark 4.8). We prove it with greedy matchings along one linear order shared by all blocks; sharing the order is what allows the blockwise choices to assemble into a global matching.

2 The reduction

Lemma 2.1. *Let D be a counterexample to Theorem 1.1 with the minimum number of vertices and, among those, the minimum number of arcs. Then $F(S) \not\subseteq S$ for every nonempty proper $S \subsetneq V(D)$; in particular every vertex of D has an in-neighbour. Moreover*

$$|F^2(S) \setminus F(S)| < |F(S) \setminus S| \qquad \text{for every nonempty proper } S \subsetneq V(D). \tag{1}$$

The arc-deletion argument below adapts the proof of [15, Lemma 4].

Proof. The vertex of a one-vertex oriented graph is good, so $|V(D)| \geq 2$. Let S be nonempty and proper with $F(S) \subseteq S$. For $v \in S$ every walk of length at most two starting at v stays in S, so $N_1^+(v)$ and $N_2^+(v)$ are the same in the induced graph $D[S]$ as in D; hence $D[S]$ is a counterexample with fewer vertices, which is impossible. For each vertex y, applying this to $S = V(D) \setminus \{y\}$, which is nonempty because $|V(D)| \geq 2$, shows that some arc enters y.

Fix a nonempty proper S and put $I = F(S) \cap S$ and $E = F(S) \setminus S$; then $E \neq \varnothing$. For $T \subseteq E$ put $g(T) = |F(T) \setminus (I \cup T)|$.

Claim. For every $T \subsetneq E$ there is a nonempty $Q \subseteq E \setminus T$ with $g(T \cup Q) - g(T) < |Q|$.

Let D_T be obtained from D by deleting all arcs from S to $E \setminus T$. Every vertex of $E \setminus T$ has an in-neighbour in S, so at least one arc is deleted, and D_T is an oriented graph on $V(D)$ with fewer arcs. By minimality some vertex v is good in D_T. If v lost no out-neighbour, then its first out-neighbourhood is unchanged and, since deleting arcs creates no walks, its second out-neighbourhood can only shrink; then v would still be bad. Hence v lost an out-neighbour, so $v \in S$, and $Q := F(\{v\}) \cap (E \setminus T)$ is nonempty: it is exactly the set of out-neighbours that v lost. Since $F(\{v\}) \subseteq F(S) = I \cup E$, the out-neighbours of v in D_T lie in $I \cup T$.

Every two-step walk $v \to u \to w$ in D_T has $u \in I \cup T$. If $u \in I$ then $u \in S$ and $w \in F(S) \setminus (E \setminus T) = I \cup T$; if $u \in T$ then $w \in F(T)$. Thus

$$F^2_{D_T}(\{v\}) \subseteq I \cup T \cup F(T). \tag{2}$$

Write $d_1 = |N_1^+(v)|$ and $d_2 = |N_2^+(v)|$ for the values in D, so $d_2 < d_1$. A vertex w that is a second out-neighbour of v in D_T but not in D lies in $F^2_D(\{v\}) \setminus \{v\}$ and outside $N_2^+(v)$, hence in $N_1^+(v)$, and it is not an out-neighbour in D_T; so $w \in Q$, and by (2) and $Q \cap (I \cup T) = \varnothing$ we get $w \in Q \cap F(T)$. Conversely let $\Lambda = F(Q) \setminus (I \cup T \cup Q \cup F(T))$. Each $w \in \Lambda$ is reached from v by a two-step walk through Q; $w \neq v$, because an arc $q \to v$ would oppose the arc $v \to q$; and $w \notin N_1^+(v) \subseteq I \cup T \cup Q$. So $\Lambda \subseteq N_2^+(v)$, while by (2) no vertex of Λ is a second out-neighbour in D_T. Goodness of v in D_T therefore gives

$$d_1 - |Q| \;\leq\; d_2 - |\Lambda| + |Q \cap F(T)|.$$

On the other hand, passing from T to $T \cup Q$ removes from the set counted by g exactly the points of $Q \cap F(T)$ and adds exactly Λ, so

$$g(T \cup Q) - g(T) = |\Lambda| - |Q \cap F(T)| \leq |Q| + d_2 - d_1 < |Q|.$$

This proves the claim.

Starting from $T = \varnothing$ and applying the claim repeatedly (each time to the original graph D) we reach $T = E$ after finitely many steps; summing the strict inequalities and using $g(\varnothing) = 0$ gives $g(E) < |E|$. Finally $F(I) \subseteq F(S) = I \cup E$, so $F^2(S) \setminus F(S) = F(I \cup E) \setminus (I \cup E) = F(E) \setminus (I \cup E)$, whose size is $g(E)$. This is (1). □

The construction in the next lemma is a lexicographic product of the form used in the proof of [1, Theorem 4.3]; here we take the second factor to be a transitive tournament. The tournament blocks magnify the subset deficit, while their internal arcs contribute only a bounded error in each fibre.

Lemma 2.2. *Let D be an oriented graph on n vertices, each with an in-neighbour, satisfying (1). Let $m > n$ and let G have vertex set $V(D) \times \{1, \dots, m\}$, with $(v, a) \to (w, b)$ if and only if $v \to w$ in D, or $v = w$ and $a < b$. Then G is an oriented graph, every vertex of G has an in-neighbour, and*

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \text{for every nonempty proper } U \subsetneq V(G). \tag{3}$$

Proof. G has no loops, and an opposite pair of arcs would require either $a < b < a$ or opposite arcs in D. An in-neighbour u of w in D gives the in-neighbour $(u, 1)$ of every (w, b).

For $\Theta \subseteq \{1, \dots, m\}$ let $f(\Theta) = \{b : \ b > a \text{ for some } a \in \Theta\}$, so $f(\varnothing) = \varnothing$. If $\Theta \neq \varnothing$ and $a = \min \Theta$, then $|f(\Theta)| = m - a$, $|f^2(\Theta)| = \max(m - a - 1, 0)$ and $|\Theta| \leq m - a + 1$, so

$$|\Theta| + |f^2(\Theta)| - 2|f(\Theta)| \leq 1. \tag{4}$$

Fix U, let $\Theta_v = \{a : (v, a) \in U\}$ and $S = \{v : \Theta_v \neq \varnothing\}$, and write $F = F_D$. The first image of U contains the full blocks over $F(S)$ and, in a block $v \in S \setminus F(S)$, exactly $\{v\} \times f(\Theta_v)$:

$$|F_G(U)| = m|F(S)| + \sum_{v \in S \setminus F(S)} |f(\Theta_v)|.$$

A two-step walk starting in U that uses two arcs between blocks ends in a block over $F^2(S)$; one using exactly one such arc ends in a block over $F(S)$; one using none stays in its block $v \in S$ and ends in $\{v\} \times f^2(\Theta_v)$. Let $H_0 = F(S) \cup F^2(S)$. The blocks over H_0 contain at most $m|H_0|$ points of $F_G^2(U)$, and every other point of $F_G^2(U)$ lies in $\{v\} \times f^2(\Theta_v)$ for some $v \in S \setminus H_0 \subseteq S \setminus F(S)$. Hence

$$|F_G^2(U)| \leq m|H_0| + \sum_{v \in S \setminus F(S)} |f^2(\Theta_v)|.$$

Using $|\Theta_v| \leq m$ for $v \in S \cap F(S)$, (4) for $v \in S \setminus F(S)$, and $|H_0| = |F(S)| + |F^2 S \setminus FS|$,

$$\begin{aligned} |U| + |F_G^2 U| - 2|F_G U| &\leq m(|S \cap F(S)| + |H_0| - 2|F(S)|) + |S \setminus F(S)| \\ &= m(|F^2 S \setminus FS| - |FS \setminus S|) + |S \setminus F(S)|. \end{aligned}$$

If $S \neq V(D)$, the bracket is a negative integer by (1), so the right side is at most $-m + n < 0$. If $S = V(D)$, positive indegrees give $F(S) = V(D)$, hence $F_G(U) = V(G)$ and then $F_G^2(U) = V(G)$, and the left side equals $|U| - |V(G)| < 0$. □

3 The extremal argument

Proof of Theorem 1.2. Suppose G is a nonempty oriented graph in which every vertex has an in-neighbour and (3) holds for every nonempty proper vertex set; we derive a contradiction. Let $X = V(G)$ and write $F = F_G$. Positive indegrees give $F(X) = X$. For $\Omega \subseteq X \times X$ define

$$F_1\Omega = \{(i,j) : \ (p,j) \in \Omega, \ p \to i\}, \qquad F_2\Omega = \{(i,j) : \ (i,s) \in \Omega, \ s \to j\},$$

so F_1 acts by F on each column $\Omega_{*j} = \{p : (p,j) \in \Omega\}$ and F_2 acts by F on each row $\Omega_{i*} = \{s : (i,s) \in \Omega\}$. Call (P,Q) *compatible* if $F_1P \cap F_2Q = \varnothing$, and let Δ be the diagonal. The pair (Δ,Δ) is compatible: $(i,j) \in F_1\Delta$ means $j \to i$ and $(i,j) \in F_2\Delta$ means $i \to j$. Among compatible pairs with $\Delta \subseteq P \cap Q$ choose one maximizing

$$\Phi(P,Q) = |P| + |Q| + d, \qquad d = |X|^2 - |F_1P| - |F_2Q|;$$

by compatibility d counts the points outside $F_1P \cup F_2Q$.

Columns and rows are proper. Each column P_{*j} contains j. If $P_{*j} = X$ then $X \times \{j\} = F(X) \times \{j\} \subseteq F_1P$; an in-neighbour w of j gives $(w,j) \in F_2\Delta \subseteq F_2Q$, contradicting compatibility. Each row Q_{i*} contains i. If $Q_{i*} = X$ then $\{i\} \times X = \{i\} \times F(X) \subseteq F_2Q$; an in-neighbour w of i gives $(i,w) \in F_1\Delta \subseteq F_1P$, again contradicting compatibility. Since $|F_1P| = \sum_j |F(P_{*j})|$, $|F_1^2P| = \sum_j |F^2(P_{*j})|$ and $|P| = \sum_j |P_{*j}|$, and likewise for Q by rows, summing (3) over the columns of P and the rows of Q gives

$$|P| + |F_1^2P| < 2|F_1P|, \qquad |Q| + |F_2^2Q| < 2|F_2Q|.$$

Let $R = X^2 \setminus F_2^2Q$ and $C = X^2 \setminus F_1^2P$. Then

$$|R| + |C| = 2|X|^2 - |F_2^2Q| - |F_1^2P| > |P| + |Q| + 2d. \tag{5}$$

We use the deletion lemma in the following special case. For $R, C \subseteq X \times X$, members $(p,j) \in R$ and $(i,s) \in C$ *conflict* if $p \to i$ and $s \to j$; a pair lying in both R and C counts as one member of each. The conflict has *Z-point* (i,j) and *H-point* (p,s), and Z and H denote the sets of all Z- and H-points. The member $(p,j) \in R$ *covers* (i,j) if $p \to i$, and $(i,s) \in C$ covers (i,j) if $s \to j$. Theorem 4.12, proved in Section 4 for general instances (here $X_1 = X_2 = X_3 = X_4 = X$ and both relations are the arc relation of G), gives $R' \subseteq R$ and $C' \subseteq C$ with no conflict between a member of R' and a member of C', containing every member that is in no conflict, such that

$$|R \setminus R'| + |C \setminus C'| \le |H| + q,$$

where q is the number of points of Z covered by no member of R' or C'. We apply it to the families R and C just defined.

The diagonal survives and has no conflicts. If $(v,v) \in F_2^2Q$, some $s \to v$ has $(v,s) \in F_2Q$; but $(s,s) \in P$ gives $(v,s) \in F_1P$. If $(v,v) \in F_1^2P$, some $p \to v$ has $(p,v) \in F_1P$; but $(p,p) \in Q$ gives $(p,v) \in F_2Q$. So $\Delta \subseteq R \cap C$. A conflict of $(v,v) \in R$ with $(i,s) \in C$ means $s \to v \to i$, and then $(s,s) \in P$ gives $(i,s) \in F_1^2P$, impossible. A conflict of $(p,j) \in R$ with $(v,v) \in C$ means $p \to v \to j$, and $(p,p) \in Q$ gives $(p,j) \in F_2^2Q$, impossible.

H-points are uncovered: $|H| \le d$. Take a conflict $(p,j) \in R$, $(i,s) \in C$. If $(p,s) \in F_1P$, some $(u,s) \in P$ has $u \to p$, and $u \to p \to i$ puts $(i,s) \in F_1^2P$. If $(p,s) \in F_2Q$, some $(p,t) \in Q$ has $t \to s$, and $t \to s \to j$ puts $(p,j) \in F_2^2Q$. Both are impossible, so H avoids $F_1P \cup F_2Q$.

Contradiction. Theorem 4.12 gives conflict-free $R' \subseteq R$, $C' \subseteq C$ containing Δ with $|R \setminus R'| + |C \setminus C'| \le |H| + q \le d + q$. A point of $F_1R' \cap F_2C'$ would come from $(p,j) \in R'$ and $(i,s) \in C'$

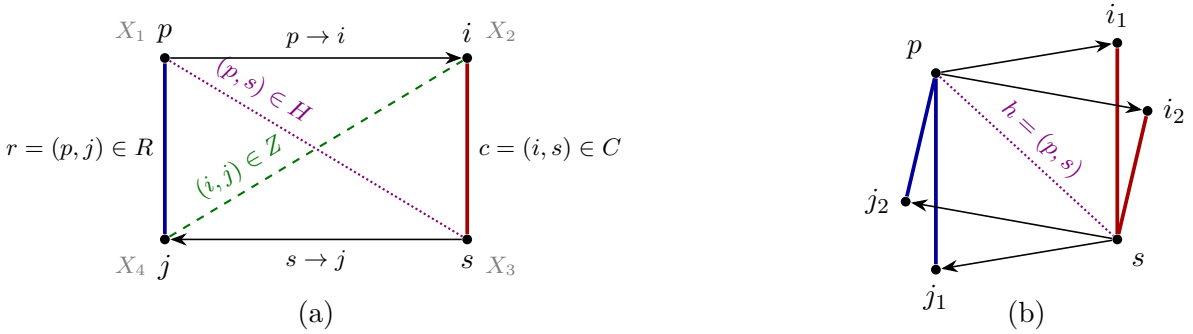


Figure 1: (a) A conflict between $r = (p, j) \in R$ and $c = (i, s) \in C$. The member r covers the Z-point (i, j) through the arc $p \to i$, and c covers it through the arc $s \to j$. The symmetry σ is the half-turn of the square: it exchanges R with C and α with β, and transposes both diagonals. (b) A block (Remark 4.2): every member (p, j_b) of $R(h)$ conflicts with every member (i_a, s) of $C(h)$; all four conflicts have the H-point $h = (p, s)$, and their Z-points (i_a, j_b) are distinct.

with $p \to i$, $s \to j$, a conflict; so (R', C') is compatible. Hence $d' := |X|^2 - |F_1R'| - |F_2C'|$ is the number of points outside $F_1R' \cup F_2C'$. A point of Z lies in $F_1R' \cup F_2C'$ exactly when it is covered by a member of R' or C', so $q \le d'$. Using (5),

$$\Phi(R', C') = |R'| + |C'| + d' \ge |R| + |C| - d - q + d' \ge |R| + |C| - d > |P| + |Q| + d = \Phi(P, Q),$$

contradicting the choice of (P, Q). □

Proof of Theorem 1.1. If the theorem failed, Lemma 2.1 would give a counterexample D on n vertices, each with an in-neighbour, satisfying (1). Lemma 2.2 with $m = n + 1$ would then give a nonempty oriented graph G in which every vertex has an in-neighbour and (3) holds for every nonempty proper vertex set, contradicting Theorem 1.2. □

4 The deletion lemma

4.1 Setting

The deletion lemma can be formulated without reference to the original oriented graph. An *instance* consists of finite sets X_1, X_2, X_3, X_4, relations $\alpha \subseteq X_1 \times X_2$ and $\beta \subseteq X_3 \times X_4$, and families $R \subseteq X_1 \times X_4$ and $C \subseteq X_2 \times X_3$. Throughout, $p \in X_1$, $i \in X_2$, $s \in X_3$, $j \in X_4$, and we write $p \to i$ for $(p, i) \in \alpha$ and $s \to j$ for $(s, j) \in \beta$. We treat R and C as two separate families: their elements are *members* (of R or of C), and a pair that lies in both R and C, which can happen when the sets X_k coincide, gives one member of each family. Members $(p, j) \in R$ and $(i, s) \in C$ *conflict* if $p \to i$ and $s \to j$; the conflict has *Z-point* (i, j) and *H-point* (p, s). Let Z and H be the sets of Z-points and H-points of all conflicts. A member $(p, j) \in R$ *covers* $(i, j) \in Z$ if $p \to i$, and a member $(i, s) \in C$ *covers* $(i, j) \in Z$ if $s \to j$. Thus a conflict is a square whose sides are the two members and the two arcs, and whose diagonals are its Z-point and its H-point (Figure 1).

Remark 4.1. (a) If $r = (p, j) \in R$ and $c = (i, s) \in C$ both cover $z = (i, j) \in Z$, they conflict, since the two covering conditions are $p \to i$ and $s \to j$. (b) A member covering z is in a conflict: a conflict with Z-point z supplies a member of the other family, and (a) applies.

For $h = (p, s) \in H$ let

$$R(h) = \{(p, j) \in R : \ s \to j\}, \qquad C(h) = \{(i, s) \in C : \ p \to i\}.$$

Remark 4.2. For $h \in H$ the conflicts with H-point h are exactly the pairs in $R(h) \times C(h)$, and both sets are nonempty. This follows directly from the definitions of conflict, $R(h)$ and $C(h)$.

The parts of every bipartite graph below are taken disjoint: when the sets involved intersect, as they do in Section 3, we use disjoint copies. Let Γ_R be the bipartite graph with parts H and R in which h is adjacent to the members of $R(h)$, and Γ_C the bipartite graph with parts H and C in which h is adjacent to the members of $C(h)$. Write ν for the maximum size of a matching. We use Hall's theorem [9] in its defect form.

Lemma 4.3. *Let ν_0 be the maximum size of a matching in a bipartite graph with parts V_1, V_2, and suppose $\nu_0 < |V_1|$. Then some nonempty $V' \subseteq V_1$ has at most $|V'| - |V_1| + \nu_0$ neighbours.*

Proof. Put $\delta = |V_1| - \nu_0 \geq 1$ and add $\delta - 1$ new vertices to V_2, each adjacent to every vertex of V_1. A matching of the new graph has at most ν_0 edges at old vertices of V_2 and at most $\delta - 1$ edges at new ones, so it does not saturate V_1. By Hall's theorem some $V' \subseteq V_1$ has fewer than $|V'|$ neighbours in the new graph; $V' \neq \varnothing$, so these neighbours include the $\delta - 1$ new vertices, and V' has fewer than $|V'| - \delta + 1$ old neighbours. □

Symmetry. For an instance $\mathcal{I} = (X_1, X_2, X_3, X_4, \alpha, \beta, R, C)$ let

$$\sigma(\mathcal{I}) = (X_3, X_4, X_1, X_2, \beta, \alpha, C^{\mathsf{T}}, R^{\mathsf{T}}), \qquad C^{\mathsf{T}} = \{(s, i) : (i, s) \in C\}, \quad R^{\mathsf{T}} = \{(j, p) : (p, j) \in R\}.$$

In Figure 1(a), σ is the half-turn of the square. Transposition identifies conflicts and coverings in $\sigma(\mathcal{I})$ with those in $\mathcal{I}$, with R and C exchanged and with Z and H transposed. It also gives $\Gamma_R(\sigma(\mathcal{I})) \cong \Gamma_C(\mathcal{I})$ and $\Gamma_C(\sigma(\mathcal{I})) \cong \Gamma_R(\mathcal{I})$; in particular, $\Gamma_L(\sigma(\mathcal{I})) \cong \Gamma_N(\mathcal{I})$ once Γ_L and Γ_N are defined below. Finally $\sigma(\sigma(\mathcal{I})) = \mathcal{I}$.

4.2 Matching numbers into X_4

For $C' \subseteq X_3$ and $X \subseteq X_4$ let $\lambda(C', X)$ be the maximum size of a matching in the bipartite graph with parts C' and X in which s is adjacent to j when $s \to j$. For $A \subseteq X_3$ let $N_\beta(A) = \{j : \ s \to j \text{ for some } s \in A\}$.

Lemma 4.4 (Kőnig)**.** $\lambda(C', X) = \min_{A \subseteq C'} \big(|C' \setminus A| + |N_\beta(A) \cap X|\big)$.

Proof. For every $A \subseteq C'$ and every matching, the edges at vertices of A end in $N_\beta(A) \cap X$ and the other edges start in $C' \setminus A$; this gives $\leq$. If $\lambda(C', X) = |C'|$, the choice $A = \varnothing$ gives equality. Otherwise Lemma 4.3 gives a nonempty $A \subseteq C'$ with $|N_\beta(A) \cap X| \leq |A| - |C'| + \lambda(C', X)$, that is, $|C' \setminus A| + |N_\beta(A) \cap X| \leq \lambda(C', X)$. □

Lemma 4.5 (bisubmodularity)**.** *For $S_1, S_2 \subseteq X_3$ and $Q_1, Q_2 \subseteq X_4$,*

$$\lambda(S_1, Q_1) + \lambda(S_2, Q_2) \ \geq \ \lambda(S_1 \cup S_2,\, Q_1 \cap Q_2) + \lambda(S_1 \cap S_2,\, Q_1 \cup Q_2).$$

Proof. Choose $A_1 \subseteq S_1$ and $A_2 \subseteq S_2$ attaining the minimum in Lemma 4.4 for (S_1, Q_1) and (S_2, Q_2). By Lemma 4.4 applied with $A_1 \cup A_2 \subseteq S_1 \cup S_2$ and with $A_1 \cap A_2 \subseteq S_1 \cap S_2$, it suffices to show

$$|(S_1 \cup S_2) \setminus (A_1 \cup A_2)| + |(S_1 \cap S_2) \setminus (A_1 \cap A_2)| = |S_1 \setminus A_1| + |S_2 \setminus A_2|,$$
$$|N_\beta(A_1 \cup A_2) \cap Q_1 \cap Q_2| + |N_\beta(A_1 \cap A_2) \cap (Q_1 \cup Q_2)| \leq |N_\beta(A_1) \cap Q_1| + |N_\beta(A_2) \cap Q_2|.$$

For the first line compare the contributions of each $s \in X_3$: if $s \in S_1 \cap S_2$, use $[s \notin A_1 \cup A_2] + [s \notin A_1 \cap A_2] = [s \notin A_1] + [s \notin A_2]$; if s lies in exactly one S_k, then s lies in neither the other set A_{3-k} nor $S_1 \cap S_2$, and both sides count it $[s \notin A_k]$ times. For the second, put $a = [j \in N_\beta(A_1)]$ and $b = [j \in N_\beta(A_2)]$. Then $[j \in N_\beta(A_1 \cup A_2)] = \max(a,b)$ and $[j \in N_\beta(A_1 \cap A_2)] \le \min(a,b)$, so j is counted on the left at most $\max(a,b) + \min(a,b) = a + b$ times if $j \in Q_1 \cap Q_2$ and at most $\min(a,b)$ times if j lies in exactly one of Q_1, Q_2; in each case this is at most its count $a[j \in Q_1] + b[j \in Q_2]$ on the right. □

Corollary 4.6. *For $S \subseteq X_3$, $X \subseteq X_4$ and $j \in X_4 \setminus X$ write $\partial_j(S,X) = \lambda(S, X \cup \{j\}) - \lambda(S,X)$.*

(a) If $X \subseteq X' \subseteq X_4$ and $j \notin X'$, then $\partial_j(S,X) \ge \partial_j(S,X')$.

(b) If $S \subseteq S' \subseteq X_3$ and $j \notin X$, then $\partial_j(S',X) \ge \partial_j(S,X)$.

(c) $\partial_j(S,X) \in \{0,1\}$ whenever $j \notin X$.

(d) $\lambda(S,X) = \lambda(S, X \cap N_\beta(S))$ for all $S \subseteq X_3$ and $X \subseteq X_4$.

Proof. (a) Lemma 4.5 with $S_1 = S_2 = S$, $Q_1 = X \cup \{j\}$, $Q_2 = X'$ gives $\lambda(S, X \cup \{j\}) + \lambda(S, X') \ge \lambda(S,X) + \lambda(S, X' \cup \{j\})$, since $(X \cup \{j\}) \cap X' = X$ and $(X \cup \{j\}) \cup X' = X' \cup \{j\}$. (b) Lemma 4.5 with $(S_1, Q_1) = (S,X)$ and $(S_2, Q_2) = (S', X \cup \{j\})$ gives $\lambda(S,X) + \lambda(S', X \cup \{j\}) \ge \lambda(S',X) + \lambda(S, X \cup \{j\})$. (c) Adding one vertex to a part changes the maximum matching size by 0 or 1. (d) Vertices of $X \setminus N_\beta(S)$ are isolated. □

Fix once and for all a linear order $\prec$ on X_4, and for $Q \subseteq X_4$ and $j \in X_4$ write $Q_{\succ j} = \{x \in Q : j \prec x\}$. For $S \subseteq X_3$ and $Q \subseteq X_4$ define the *greedy set*

$$g(S,Q) = \{j \in Q : \ \partial_j(S, Q_{\succ j}) = 1\}.$$

Adding the points of Q in decreasing order changes λ by 0 or 1 at each step; the greedy set consists precisely of the points giving an increment of 1.

Lemma 4.7. *$|g(S,Q)| = \lambda(S,Q)$, and every $Q_0 \subseteq g(S,Q)$ satisfies $\lambda(S,Q_0) = |Q_0|$.*

Proof. The first assertion is the telescoping identity $\lambda(S,Q) = \sum_{j \in Q} \partial_j(S, Q_{\succ j})$ together with Corollary 4.6(c). Let $G = g(S,Q)$. For $j \in G$ we have $G_{\succ j} \subseteq Q_{\succ j}$ and $j \notin Q_{\succ j}$, so Corollary 4.6(a) gives $\partial_j(S, G_{\succ j}) \ge \partial_j(S, Q_{\succ j}) = 1$. Summing these increments along G gives $\lambda(S,G) \ge |G|$; since trivially $\lambda(S,G) \le |G|$, we get $\lambda(S,G) = |G|$. Thus G, and hence every $Q_0 \subseteq G$, can be matched into S. □

4.3 The counting lemma

Let Γ_L be the bipartite graph with parts R and Z in which a member (p,j) is adjacent to the points of Z that it covers, and Γ_N the bipartite graph with parts C and Z in which a member (i,s) is adjacent to the points of Z that it covers. By the symmetry paragraph, $\Gamma_L(\sigma(\mathcal{I})) \cong \Gamma_N(\mathcal{I})$.

Remark 4.8. Although the proof below is purely combinatorial, Lemma 4.9 has a useful linear-algebraic interpretation. Take one indeterminate x_{pi} for each $p \to i$ and one indeterminate y_{sj} for each $s \to j$, and let $L \in \mathbb{K}^{Z \times R}$, $N \in \mathbb{K}^{C \times Z}$, $B \in \mathbb{K}^{H \times R}$ and $A \in \mathbb{K}^{C \times H}$, over the field $\mathbb{K}$ of rational functions in these indeterminates, have the entries $L_{(i,j),(p,j)} = x_{pi}$, $N_{(i,s),(i,j)} = y_{sj}$, $B_{(p,s),(p,j)} = y_{sj}$ and $A_{(i,s),(p,s)} = x_{pi}$ when $p \to i$, respectively $s \to j$, and 0 otherwise. Both $(NL)_{(i,s),(p,j)}$ and $(AB)_{(i,s),(p,j)}$ equal $x_{pi}y_{sj}$ if $p \to i$ and $s \to j$ (then (p,j) and (i,s) conflict,

so $(i,j) \in Z$ and $(p,s) \in H$) and 0 otherwise; so $NL = AB$. The matrices L, N and B are block diagonal (over j, i and p respectively) with distinct indeterminates within each block, so by [4] their ranks are $\nu(\Gamma_L)$, $\nu(\Gamma_N)$ and $\nu(\Gamma_R)$. Sylvester's inequality and $\operatorname{rank}(AB) \le \operatorname{rank} B$ give $\operatorname{rank} L + \operatorname{rank} N - |Z| \le \operatorname{rank}(NL) = \operatorname{rank}(AB) \le \operatorname{rank} B$, which is the first inequality of Lemma 4.9. This interpretation is not used in the combinatorial proof below.

Lemma 4.9. *For every instance, $\nu(\Gamma_L) + \nu(\Gamma_N) \le |Z| + \nu(\Gamma_R) \le |Z| + |H|$.*

In the proof, the greedy sets B_i of the blocks C_i are all taken along the order $\prec$. Corollary 4.6 then shows that they remain greedy for the larger sets $C^{(p)}$ (Step 2), and this is where the common order is used.

Proof. For $i \in X_2$ let $C_i = \{s : (i,s) \in C\}$ and $Z^i = \{j : (i,j) \in Z\}$; for $p \in X_1$ let $R_p = \{j : (p,j) \in R\}$ and $C^{(p)} = \bigcup_{i:\, p \to i} C_i$. Two observations:

(A) If $p \to i$, then $R_p \cap N_\beta(C_i) \subseteq Z^i$: if $j \in R_p$ and $s \in C_i$ with $s \to j$, then (p,j) and (i,s) conflict, with Z-point (i,j).

(B) Γ_N is the disjoint union over i of the graphs between the members (i,s), $s \in C_i$, and the points (i,j), $j \in Z^i$, with (i,s) adjacent to (i,j) when $s \to j$. Hence $\nu(\Gamma_N) = \sum_i \lambda(C_i, Z^i)$.

Let $B_i = g(C_i, Z^i) \subseteq Z^i$ and $Y = \{(i,j) :\ j \in B_i\} \subseteq Z$. By Lemma 4.7 and (B), $|Y| = \nu(\Gamma_N)$. Let W be the set of $(p,j) \in R$ for which $(i,j) \in Y$ for some i with $p \to i$.

Step 1: $\nu(\Gamma_L) \le |W| + |Z| - |Y|$. A member $(p,j) \in R \setminus W$ is adjacent in Γ_L only to points (i,j) with $p \to i$, and these are not in Y. So in a matching of Γ_L the edges at $R \setminus W$ end at distinct points of $Z \setminus Y$, and there are at most $|W|$ other edges.

Step 2: W can be matched into H in Γ_R. Fix p and let $W_p = \{j : (p,j) \in W\} \subseteq R_p$. Let $j \in W_p$, choose i with $p \to i$ and $j \in B_i$, and put $X = (R_p)_{\succ j}$. Since $j \in B_i$, $\partial_j(C_i, Z^i_{\succ j}) = 1$. By (A), $X \cap N_\beta(C_i) \subseteq Z^i_{\succ j}$, so Corollary 4.6(a) gives $\partial_j(C_i, X \cap N_\beta(C_i)) \ge 1$. Since $j \in Z^i$, some $s \in C_i$ has $s \to j$, so $j \in N_\beta(C_i)$ and $(X \cup \{j\}) \cap N_\beta(C_i) = (X \cap N_\beta(C_i)) \cup \{j\}$; applying Corollary 4.6(d) to X and to $X \cup \{j\}$ gives $\partial_j(C_i, X) = \partial_j(C_i, X \cap N_\beta(C_i)) \ge 1$. As $C_i \subseteq C^{(p)}$, Corollary 4.6(b) gives $\partial_j(C^{(p)}, X) \ge 1$, hence $\partial_j(C^{(p)}, (R_p)_{\succ j}) = 1$ by Corollary 4.6(c); that is, $j \in g(C^{(p)}, R_p)$. Thus $W_p \subseteq g(C^{(p)}, R_p)$, and Lemma 4.7 gives an injective map $m_p \colon W_p \to C^{(p)}$ with $m_p(j) \to j$ for all j. For $j \in W_p$ the element $s = m_p(j)$ lies in $C_{i'}$ for some i' with $p \to i'$; then (p,j) and (i',s) conflict, so $(p,s) \in H$ and $(p,j) \in R(p,s)$. The edges $(p,j) \mapsto (p, m_p(j))$, over all p and $j \in W_p$, form a matching of Γ_R that covers W: within one p the points $(p, m_p(j))$ are distinct because m_p is injective, and for different p they are distinct because their first coordinates differ.

By Steps 1 and 2, $\nu(\Gamma_L) \le \nu(\Gamma_R) + |Z| - \nu(\Gamma_N)$; and $\nu(\Gamma_R) \le |H|$ because H is one part of Γ_R. □

The next lemma is the only place where we use the hypotheses of the counting lemma.

Lemma 4.10 (Local lemma)**.** *If every member of R is in a conflict and Γ_C has a matching saturating H, then Γ_L has a matching saturating R.*

Proof. Γ_L is the disjoint union, over $j \in X_4$, of its subgraphs Γ^j_L on the members $(p,j) \in R$ and the points $(i,j) \in Z$. Fix j and write N_L for neighbourhoods in Γ^j_L; by Hall's theorem it suffices to show that every set U of members $(p,j) \in R$ has at least $|U|$ neighbours. Suppose not, and choose such a U with $|N_L(U)| < |U|$ and $|U|$ minimal; put $T = N_L(U)$. For every $w \in U$ the set $U \setminus \{w\}$ satisfies Hall's condition, by minimality, so it has a matching into T saturating it; hence $|T| = |U| - 1$, and this matching is a bijection $\mu_w \colon U \setminus \{w\} \to T$ with $\mu_w(r) \in N_L(r)$ for every r.

Let $S_j = \{s : s \to j\}$. For $s \in S_j$ let U_s be the set of $(p, j) \in U$ for which some $(i, s) \in C$ has $p \to i$, and let T_s be the set of $(i, j) \in T$ with $(i, s) \in C$. If $(p, j) \in U_s$ and $(i, s) \in C$ with $p \to i$, then (p, j) and (i, s) conflict because $s \to j$; so $(p, s) \in H$ and $(i, j) \in Z$, and (i, j) is covered by (p, j), so $(i, j) \in T_s$. A matching of Γ_C saturating H sends the distinct H-points (p, s), $(p, j) \in U_s$, to distinct members $(i, s) \in C$ with $p \to i$, hence to distinct points $(i, j) \in T_s$. Therefore $|U_s| \le |T_s|$ for every $s \in S_j$, and summing,

$$\sum_{r \in U} m_r \le \sum_{z \in T} n_z, \qquad m_r = \#\{s \in S_j : r \in U_s\}, \quad n_z = \#\{s \in S_j : z \in T_s\}. \tag{6}$$

If $r = (p, j) \in U$ is adjacent to $z = (i, j)$ in Γ_L, then $p \to i$, so every $s \in S_j$ with $z \in T_s$ has $(i, s) \in C$ and hence $r \in U_s$; thus $m_r \ge n_z$. Fix $w \in U$. Using μ_w,

$$\sum_{r \in U} m_r \;\ge\; m_w + \sum_{r \in U \setminus \{w\}} n_{\mu_w(r)} \;=\; m_w + \sum_{z \in T} n_z.$$

Finally $m_w \ge 1$: by hypothesis $w = (p, j)$ conflicts with some $(i, s) \in C$, and then $s \in S_j$ and $w \in U_s$. This contradicts (6). □

Lemma 4.11 (Counting lemma)**.** *Suppose that every member is in a conflict and that both Γ_R and Γ_C have matchings saturating H. Then $|R| + |C| \le |Z| + |H|$.*

Proof. By Lemma 4.10, $\nu(\Gamma_L) = |R|$. The instance $\sigma(\mathcal{I})$ also has every member in a conflict, and $\Gamma_C(\sigma(\mathcal{I})) \cong \Gamma_R(\mathcal{I})$ has a matching saturating its H-points; so Lemma 4.10 applied to $\sigma(\mathcal{I})$ gives $\nu(\Gamma_N(\mathcal{I})) = \nu(\Gamma_L(\sigma(\mathcal{I}))) = |C|$. Lemma 4.9 concludes. □

4.4 The deletion lemma

Theorem 4.12 (Deletion lemma)**.** *There are $R' \subseteq R$ and $C' \subseteq C$ with no conflict between a member of R' and a member of C', containing every member that is in no conflict, such that*

$$|R \setminus R'| + |C \setminus C'| \le \min\{\nu(\Gamma_R), \nu(\Gamma_C)\} + q,$$

where q is the number of points of Z covered by no member of R' or C'. In particular $|R \setminus R'| + |C \setminus C'| \le |H| + q$.

The proof treats members without conflicts (Case 1), a Hall deficiency on one side (Case 2, where σ reduces the side of C to the side of R), and the saturated case (Case 3).

Proof. Call a pair (R', C') with the stated properties a *solution*. We use induction on $|R| + |C|$; the induction hypothesis covers every instance, of any shape, with fewer members. If there is no conflict, then $\nu(\Gamma_R) = \nu(\Gamma_C) = 0$ and (R, C) is a solution. Let $\mathfrak{m} = \min\{\nu(\Gamma_R), \nu(\Gamma_C)\}$.

Case 1: some member e is in no conflict. Removing e from its family changes neither the conflicts nor Z nor H. The member e is isolated in its corresponding graph, while the other graph is unchanged, so $\mathfrak{m}$ is unchanged. Apply the induction hypothesis to the families without e and add e back to the kept family it came from. By Remark 4.1(b), e covers no point of Z, so this changes neither the deletion count nor q, and creates no conflict.

Case 2: every member is in a conflict and $\mathfrak{m} < |H|$. Assume first that $\nu(\Gamma_R) = \mathfrak{m}$. By Lemma 4.3 there is a nonempty $H' \subseteq H$ whose set of neighbours $W = \bigcup_{h \in H'} R(h)$ in Γ_R satisfies $|W| \le |H'| - |H| + \mathfrak{m}$; and $W \neq \varnothing$ by Remark 4.2. Apply the induction hypothesis to the families

$R \setminus W$ and C; write Z°, H° for their sets of Z- and H-points, and q° for their count of uncovered points. It gives $R' \subseteq R \setminus W$ and $C' \subseteq C$ with no conflict between them and

$$|(R \setminus W) \setminus R'| + |C \setminus C'| \le \min\{\nu(\Gamma_R^\circ), \nu(\Gamma_C^\circ)\} + q^\circ \le |H^\circ| + q^\circ.$$

Every conflict among $R \setminus W$ and C is a conflict among R and C, so $Z^\circ \subseteq Z$ and $H^\circ \subseteq H$. No $h \in H'$ lies in H°, because by Remark 4.2 every conflict with H-point h involves a member of $R(h) \subseteq W$; hence $|H^\circ| \le |H| - |H'|$. A point of Z° covered by no member of R' or C' is a point of Z covered by no member of R' or C', so $q^\circ \le q$. Since every member is in a conflict, the requirement on members without conflicts is empty, and (R', C') is a solution:

$$|R \setminus R'| + |C \setminus C'| \le |W| + |H| - |H'| + q \le \mathfrak{m} + q.$$

If instead $\nu(\Gamma_C) = \mathfrak{m} < \nu(\Gamma_R)$, apply the previous paragraph to $\sigma(\mathcal{I})$, which has the same number of members, satisfies the hypotheses of Case 2 with $\nu(\Gamma_R(\sigma(\mathcal{I}))) = \nu(\Gamma_C(\mathcal{I})) = \mathfrak{m}$, and only invokes the induction hypothesis on instances with fewer members. By the identification of $\sigma(\mathcal{I})$ with $\mathcal{I}$ described in the symmetry paragraph, a solution (R'', C'') for $\sigma(\mathcal{I})$ gives the solution $((C'')^\mathsf{T}, (R'')^\mathsf{T})$ for $\mathcal{I}$, with the same deletion count and the same q.

Case 3: every member is in a conflict and $\mathfrak{m} = |H|$, so both Γ_R and Γ_C have matchings saturating H. Take $R' = C' = \varnothing$. There is no conflict between empty families, there is no member without a conflict, and every point of Z is uncovered, so $q = |Z|$. By Lemma 4.11, $|R| + |C| \le |H| + |Z| = \mathfrak{m} + q$. □

Remark 4.13. Let Γ be the bipartite graph used in [13]: its left part is R together with a copy Z_L of Z, its right part is C together with a copy Z_R of Z, and its edges join conflicting members, r to z_R when $r \in R$ covers z, and z_L to c when $c \in C$ covers z. Theorem 4.12 gives $\nu(\Gamma) \le |Z| + \min\{\nu(\Gamma_R), \nu(\Gamma_C)\}$, which sharpens the bound $\nu(\Gamma) \le |Z| + |H|$ of [13]. Indeed, for a solution (R', C') let K consist of the deleted members, the copies z_R of the points covered by members of R', and the copies z_L of the points covered by members of C'. By Remark 4.1(a) no point is covered by members of both R' and C', so $|K| = |R \setminus R'| + |C \setminus C'| + |Z| - q$. Every edge of Γ meets K: a conflict has a deleted endpoint, and an edge from a kept member to a copy of z ends in K by construction. A matching meets each vertex of K at most once, so $\nu(\Gamma) \le |K|$.

5 Two consequences

Corollary 5.1 (weighted form)**.** *Let D be a nonempty oriented graph and $w\colon V(D) \to [0, \infty)$. Then some vertex v satisfies $w(N_1^+(v)) \le w(N_2^+(v))$, where $w(S) = \sum_{x \in S} w(x)$.*

For the related arc-weighted formulation and its equivalence with the unweighted conjecture, see Seacrest [14].

Proof. First let w take positive integer values. Let D_w have vertex set $\{(v, k) : v \in V(D),\ 1 \le k \le w(v)\}$ and arcs $(v, k) \to (u, l)$ whenever $v \to u$. It is a nonempty oriented graph. For $x = (v, k)$ we have $N_1^+(x) = \{(u, l) : u \in N_1^+(v)\}$. A vertex (u, l) is reached from x by a two-step walk exactly when $u \in F^2(\{v\})$; it is not x and not an out-neighbour of x exactly when $u \notin N_1^+(v)$, because $u = v$ would need a closed walk of length two in D. So $N_2^+(x) = \{(u, l) : u \in N_2^+(v)\}$, and a good vertex of D_w, given by Theorem 1.1, yields the required v. Scaling handles positive rational weights. For each v the set of $w \in [0, \infty)^{V(D)}$ with $w(N_1^+(v)) \le w(N_2^+(v))$ is closed, so their union is closed; it contains the dense set of positive rational weight vectors, hence all of $[0, \infty)^{V(D)}$. □

Corollary 5.2 (directed triangles)**.** *If an oriented graph D on $n \geq 1$ vertices has no directed triangle, then some vertex v satisfies $2|N_1^+(v)| + |N_1^-(v)| \leq n - 1$, where $N_1^-(v) = \{u : u \to v\}$. Consequently every oriented graph on $n \geq 1$ vertices in which every vertex has at least $n/3$ out-neighbours and at least $n/3$ in-neighbours contains a directed triangle.*

Proof. Let v be good. If some $w \in N_2^+(v)$ were an in-neighbour of v, a walk $v \to u \to w$ would close to the directed triangle $v \to u \to w \to v$. Hence the sets $\{v\}$, $N_1^+(v)$, $N_1^-(v)$ and $N_2^+(v)$ are pairwise disjoint ($N_1^+(v) \cap N_1^-(v) = \varnothing$ because D is oriented), and $|N_1^+(v)| \leq |N_2^+(v)| \leq n - 1 - |N_1^+(v)| - |N_1^-(v)|$, which is the first assertion. Under the degree condition every vertex has $2|N_1^+(v)| + |N_1^-(v)| \geq n > n - 1$, so the graph cannot be free of directed triangles. □

6 Formal verification

All numbered lemmas, theorems and corollaries of Sections 1–5 have been checked in Lean 4 (version 4.36.0-rc2) with Mathlib (tag v4.36.0-rc2). The source files accompany the paper as the library `Seymour`. The remarks are not formalized, although the facts in Remarks 4.1 and 4.2 are proved wherever the argument uses them.

The formal statement of Theorem 1.1 uses no definitions from the development: for every nonempty finite type X and every relation E on X with $E\,x\,y \Rightarrow \neg E\,y\,x$, some v satisfies

$$|\{u : E\,v\,u\}| \leq |\{u : \neg E\,v\,u \text{ and } E\,v\,w,\ E\,w\,u \text{ for some } w\}|.$$

The command `#print axioms` reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`, and the library builds without `sorry` or warnings.

Section 4 is formalized for general instances. The implementation conventions—a common universe for the four types, direct definitions of matching numbers, an enumerated linear order and truncated subtraction for increments—do not change the mathematical statements.

A traversal of the proof terms finds no constants from matrices, linear maps, submodules, ranks, polynomial rings or fraction fields.

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